Cairo University

Faculty of Engineering

Aerospace Department

Assignment 3

Finite Element Method in Fluid Mechanics

Numerical Integration, Shape function

Name	Student Code
Mohamed Abd El-Mawgoud Ghoneam	202211458

Submitted to:

Dr. Amr Guaily

March 13, 2026

Question (1)

$$\int_a^b f(x)\,dx$$

Using the Gauss-Legendre transformation

$$x = \frac{a+b}{2} + \frac{b-a}{2}\xi$$

$$dx = \frac{b-a}{2}d\xi$$

Therefore

$$I = \frac{b-a}{2}\int_{-1}^{1} f\left(\frac{a+b}{2} + \frac{b-a}{2}\xi\right)d\xi$$

Integral I_1

$$I_1 = \int_{-3}^{9} x^2 dx$$

From the transformation

$$x = \frac{9+(-3)}{2} + \frac{9-(-3)}{2}\xi$$

$$x = 3 + 6\xi$$

and

$$dx = 6d\xi$$

So

$$I_1 = \int_{-1}^{1} (3+6\xi)^2 6d\xi$$

Integral I_2

$$I_2 = \int_{-\pi}^{3\pi} \cos(x)dx$$

Transformation

$$x = \frac{-\pi+3\pi}{2} + \frac{3\pi-(-\pi)}{2}\xi$$

$$x = \pi + 2\pi\xi$$

Thus

$$I_2 = \int_{-1}^{1} \cos(\pi + 2\pi\xi) 2\pi d\xi$$

Integral I_3

$$I_3 = \int_{-1}^{1} \int_{-1}^{1} (\xi + \eta) d\xi d\eta$$

Gauss-Legendre formula

$$I = \sum_{i=1}^{n} G(\xi_i) W_i$$

2-Point Formula ($n = 2$)

Nodes

$$\xi_i = \pm \frac{1}{\sqrt{3}}$$

Weights

$$W_i = 1$$

For I_1

$$I_1 = 6[(3 + 6\frac{1}{\sqrt{3}})^2 + (3 - 6\frac{1}{\sqrt{3}})^2]$$

$$I_1 = 6[(3 + \sqrt{12})^2 + (3 - \sqrt{12})^2]$$

$$I_1 = 252$$

For I_2

$$I_2 = 2\pi[\cos(\pi - \frac{2\pi}{\sqrt{3}}) + \cos(\pi + \frac{2\pi}{\sqrt{3}})]$$

$$I_2 = 2\pi[-2\cos(\frac{2\pi}{\sqrt{3}})]$$

$$I_2 = -11.11255362$$

For I_3

$$I_3 = \sum_{i=1}^{n} \sum_{j=1}^{n} w_i w_j (\xi_i + \eta_j)$$

Since the nodes are symmetric

$$\xi_1 + \xi_2 = 0$$

Therefore

$$I_3 = 0$$

3-Point Formula ($n = 3$)

Nodes

$$\xi_1 = 0$$

$$\xi_{2,3} = \pm\sqrt{\frac{3}{5}} = \pm\sqrt{0.6}$$

Weights

$$W_1 = \frac{8}{9}$$

$$W_2 = W_3 = \frac{5}{9}$$

For I_1

$$I_1 = 6\left[\frac{8}{9}(3)^2 + \frac{5}{9}(3 + 6\sqrt{0.6})^2 + \frac{5}{9}(3 - 6\sqrt{0.6})^2\right]$$

$$I_1 = 252$$

For I_2

$$I_2 = 2\pi\left[\frac{8}{9}\cos(\pi) + \frac{5}{9}\cos(\pi + 2\pi\sqrt{0.6}) + \frac{5}{9}\cos(\pi - 2\pi\sqrt{0.6})\right]$$

$$I_2 = -6.65969$$

For I_3

$$I_3 = \frac{8}{9}(0) + \frac{5}{9}\sqrt{0.6} + \frac{5}{9}(-\sqrt{0.6})$$

$$I_3 = 0$$

Exact solution

I_1

$$I_1 = \int_{-3}^{9} x^2 dx$$

$$= \frac{x^3}{3}\Big|_{-3}^{9}$$

$$= 252$$

I_2

$$I_2 = \int_{-\pi}^{3\pi} \cos(x) dx$$

$$= [\sin(x)]_{-\pi}^{3\pi}$$

$$= 0$$

I_3

$$I_3 = \int_{-1}^{1}\int_{-1}^{1} (\xi + \eta) d\xi d\eta$$

$$= \int_{-1}^{1} \left[\frac{\xi^2}{2} + \eta\xi\right]_{-1}^{1} d\eta$$

$$= \int_{-1}^{1} (0 + 2\eta) d\eta$$

$$= 0$$

Comparison of Results

Integral	Exact Value	2-Point Result	3-Point Result	Error
$I_1 = \int_{-3}^{9} x^2 dx$	252	252	252	(0% error)
$I_2 = \int_{-\pi}^{3\pi} \cos(x) dx$	0	-11.1126	-6.6597	Large error
$I_3 = \int_{-1}^{1} \int_{-1}^{1} (\xi + \eta) d\xi d\eta$	0	0	0	(0% error)

Discussion

Observation	Explanation
I_1	The integrand x^2 is a polynomial of degree 2, and Gauss–Legendre with $n = 2$ integrates polynomials up to degree $2n - 1 = 3$ exactly.
I_2	The function $\cos(x)$ is an even function and it is not a polynomial and oscillates over the interval, so the approximation using a small number of Gauss points produces noticeable error.
I_3	The integrand is symmetric over the interval $[-1, 1]$, causing the positive and negative contributions to cancel and giving the exact result.

Question (2)

For the quadrilateral element, the corner nodes Cartesian locations are

$$[(0.2, 0.2), (0.4, 0.4), (0.35, 0.5), (0.1, 0.6)]$$

Shape Functions

The shape functions for a four-node quadrilateral element can be written as

$$N_i(\xi, \eta) = a + b\xi + c\eta + d\xi\eta$$

where

$$N_i = \begin{cases} 1 & \text{at node } i \\ 0 & \text{otherwise} \end{cases}$$

The general form of interpolation is

$$S = A + B\xi + C\eta + D\xi\eta$$

Using the interpolation conditions for each node, the shape functions are obtained as follows

$$N_1 = \frac{(\xi - 1)(\eta - 1)}{(\xi_1 - 1)(\eta_1 - 1)} = \frac{1}{4}(1 - \xi)(1 - \eta)$$

$$N_2 = \frac{(\xi - (-1))(\eta - 1)}{(\xi_2 - (-1))(\eta_2 - 1)} = \frac{1}{4}(1 + \xi)(1 - \eta)$$

$$N_3 = \frac{(\xi - (-1))(\eta - (-1))}{(\xi_3 - (-1))(\eta_3 - (-1))} = \frac{1}{4}(1 + \xi)(1 + \eta)$$

$$N_4 = \frac{(\xi - 1)(\eta - (-1))}{(\xi_4 - 1)(\eta_4 - (-1))} = \frac{1}{4}(1 - \xi)(1 + \eta)$$

(a) Calculation of x and y

The coordinate transformation is

$$x(\xi,\eta) = \sum N_i x_i$$

$$y(\xi,\eta) = \sum N_i y_i$$

Thus

$$x(\xi,\eta) = 0.05(1-\xi)(1-\eta) + 0.1(1+\xi)(1-\eta) + 0.0875(1+\xi)(1+\eta) + 0.025(1-\xi)(1+\eta)$$

$$y(\xi,\eta) = 0.05(1-\xi)(1-\eta) + 0.1(1+\xi)(1-\eta) + 0.125(1+\xi)(1+\eta) + 0.15(1-\xi)(1+\eta)$$

Evaluating at

$$(\xi,\eta) = (0.5, 0.5)$$

$$x(0.5, 0.5) = 0.303125$$

$$y(0.5, 0.5) = 0.48125$$

(b) Evaluation of the derivatives of Shape Functions N_{1x} and N_{1y}

Using the chain rule

$$\frac{\partial N_i}{\partial \xi} = \frac{\partial N_i}{\partial x}\frac{\partial x}{\partial \xi} + \frac{\partial N_i}{\partial y}\frac{\partial y}{\partial \xi}$$

$$\frac{\partial N_i}{\partial \eta} = \frac{\partial N_i}{\partial x}\frac{\partial x}{\partial \eta} + \frac{\partial N_i}{\partial y}\frac{\partial y}{\partial \eta}$$

Using the Jacobian transformation

$$\begin{bmatrix} N_{i\xi} \\ \\ N_{i\eta} \end{bmatrix} = \begin{bmatrix} x_\xi & y_\xi \\ \\ x_\eta & y_\eta \end{bmatrix} \begin{bmatrix} N_{ix} \\ \\ N_{iy} \end{bmatrix}$$

Therefore

$$\begin{bmatrix} N_{ix} \\ \\ N_{iy} \end{bmatrix} = \frac{1}{x_\xi y_\eta - x_\eta y_\xi} \begin{bmatrix} y_\eta & -y_\xi \\ \\ -x_\eta & x_\xi \end{bmatrix} \begin{bmatrix} N_{i\xi} \\ \\ N_{i\eta} \end{bmatrix}$$

Thus

$$N_{ix} = \frac{y_\eta N_{i\xi} - y_\xi N_{i\eta}}{x_\xi y_\eta - x_\eta y_\xi}$$

$$N_{iy} = \frac{x_\xi N_{i\eta} - x_\eta N_{i\xi}}{x_\xi y_\eta - x_\eta y_\xi}$$

The derivatives with respect to ξ and η are

$$\frac{\partial N_1}{\partial \xi} = \frac{1}{4}(\eta - 1) \qquad \frac{\partial N_1}{\partial \eta} = \frac{1}{4}(\xi - 1)$$

$$\frac{\partial N_2}{\partial \xi} = \frac{1}{4}(1 - \eta) \qquad \frac{\partial N_2}{\partial \eta} = -\frac{1}{4}(1 + \xi)$$

$$\frac{\partial N_3}{\partial \xi} = \frac{1}{4}(1 + \eta) \qquad \frac{\partial N_3}{\partial \eta} = \frac{1}{4}(1 + \xi)$$

$$\frac{\partial N_4}{\partial \xi} = -\frac{1}{4}(1 + \eta) \qquad \frac{\partial N_4}{\partial \eta} = \frac{1}{4}(1 - \xi)$$

The derivatives of the coordinates are

$$x_\xi = N_{i\xi} x_i = 0.05(\eta - 1) + 0.1(1 - \eta) + 0.0875(1 + \eta) - 0.025(1 + \eta) = 0.0125\eta + 0.1125$$

$$x_\eta = N_{i\eta} x_i = 0.05(\xi - 1) + 0.1(1 + \xi) + 0.0875(1 + \xi) - 0.025(1 - \xi) = 0.0125\xi - 0.0375$$

$$y_\xi = N_{i\xi} y_i = 0.05(\eta - 1) + 0.1(1 - \eta) + 0.125(1 + \eta) - 0.15(1 + \eta) = -0.075\eta + 0.025$$

$$y_\eta = N_{i\eta} y_i = 0.05(\xi - 1) - 0.1(1 + \xi) + 0.125(1 + \xi) + 0.15(1 - \xi) = -0.075\xi + 0.125$$

$$|J| = x_\xi y_\eta - y_\xi x_\eta = (0.0125\eta + 0.1125)(-0.075\xi + 0.125) - (0.0125\xi - 0.0375)(-0.075\eta + 0.025)$$

therefore

$$N_{1x} = \frac{0.25\left[(-0.075\xi + 0.125)(\eta - 1) + (-0.075\eta + 0.025)(\xi - 1)\right]}{|J|}$$

$$N_{1y} = \frac{0.25\left[(0.0125\eta + 0.1125)(\xi - 1) - (0.0125\xi - 0.0375)(\eta - 1)\right]}{|J|}$$

Finally, evaluating at

$$(\xi, \eta) = (0.5, 0.5)$$

$$N_{1x} = -1.25$$

$$N_{1y} = -1.875$$

Part (c): Evaluation of the Element Stiffness Matrix

The element stiffness matrix for a four–node quadrilateral element is defined as

$$k_{ij} = \iint_{A_e} \left(\frac{\partial N_i}{\partial x}\frac{\partial N_j}{\partial x} + \frac{\partial N_i}{\partial y}\frac{\partial N_j}{\partial y} \right) dx\, dy \tag{1}$$

Since the element geometry is irregular, the integration is transformed from the physical coordinates (x, y) to the natural coordinates (ξ, η).

Coordinate Transformation

Using the isoparametric mapping

$$x = \sum_{i=1}^{4} N_i x_i, \qquad y = \sum_{i=1}^{4} N_i y_i \tag{2}$$

the area integral becomes

$$k_{ij} = \int_{-1}^{1}\int_{-1}^{1} \left(\frac{\partial N_i}{\partial x}\frac{\partial N_j}{\partial x} + \frac{\partial N_i}{\partial y}\frac{\partial N_j}{\partial y} \right) |J|\, d\xi d\eta \tag{3}$$

where J is the Jacobian matrix

$$J = \begin{bmatrix} \dfrac{\partial x}{\partial \xi} & \dfrac{\partial x}{\partial \eta} \\[2ex] \dfrac{\partial y}{\partial \xi} & \dfrac{\partial y}{\partial \eta} \end{bmatrix} \tag{4}$$

and $|J|$ is its determinant.

Transformation of Derivatives

The derivatives of the shape functions with respect to the physical coordinates are obtained using

$$\begin{bmatrix} \dfrac{\partial N_i}{\partial x} \\[2ex] \dfrac{\partial N_i}{\partial y} \end{bmatrix} = J^{-1} \begin{bmatrix} \dfrac{\partial N_i}{\partial \xi} \\[2ex] \dfrac{\partial N_i}{\partial \eta} \end{bmatrix} \tag{5}$$

Numerical Integration

The integral is evaluated using the 2×2 Gauss quadrature rule.

The Gauss points are

$$\xi, \eta = \pm \frac{1}{\sqrt{3}} \tag{6}$$

with corresponding weights

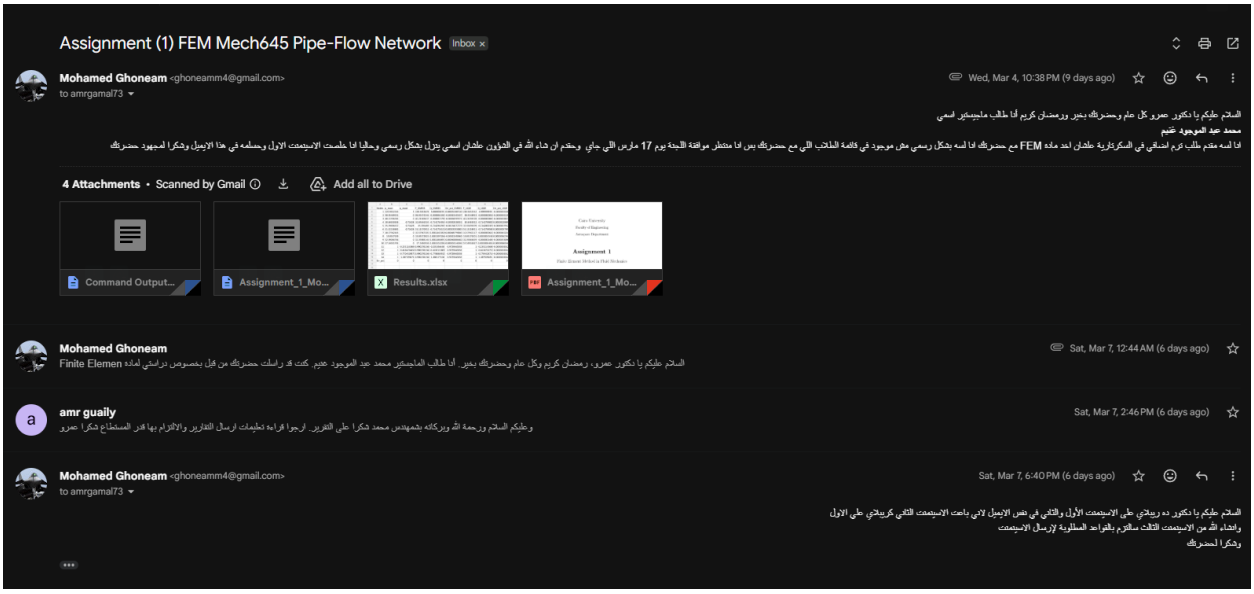
$$w_1 = w_2 = 1 \tag{7}$$

Thus the stiffness matrix becomes

$$k_{ij} = \sum_{m=1}^{2} \sum_{n=1}^{2} \left(\frac{\partial N_i}{\partial x} \frac{\partial N_j}{\partial x} + \frac{\partial N_i}{\partial y} \frac{\partial N_j}{\partial y} \right) |J| w_m w_n \tag{8}$$

Final Element Stiffness Matrix

After evaluating the integrals using Matlab R2018b at the four Gauss points, the element stiffness matrix is obtained as

$$k = \begin{bmatrix} 0.5629 & -0.1581 & -0.4354 & 0.0306 \\ -0.1581 & 0.8990 & -0.4155 & -0.3253 \\ -0.4354 & -0.4155 & 1.0728 & -0.2218 \\ 0.0306 & -0.3253 & -0.2218 & 0.5166 \end{bmatrix}$$

السلام عليكم يا دكتور عمرو انا بفكر حضرتك اني ارسلت اسينمنت 1 وحضرتك طلبت مني ابعت ريبلاي علي الميل الي باعته وبالفعل عملت المطلوب في نفس اليوم وانا ارسلته متاخرا بسبب اني كنت مقدم ترم اضافي متاخرا كما وضحت سابقا وهذه صورة للميل للاسينمنت الاول